

Financial Mathematics: Exercises Lecture 4: Stochastic Calculus

$(W_t)_{t \geq 0}$ standard Brownian motion

Exercise 1

Let X_t satisfy the SDE

$$dX_t = \mu dt + \sigma dW_t.$$

Apply Itô's formula to compute the SDE satisfied by:

1. X_t
2. e^{X_t}
3. $\sin(X_t)$.

1. $f(x) = x^2$.

Itô formula for an Itô process:

$$(*) df(t, X(t)) = f_t(t, X(t))dt + f_x(t, X(t))dX(t) + \frac{1}{2}f_{xx}(t, X(t))dX(t)dX(t)$$

$$f_t(x) = 0, f_x(x) = 2x, f_{xx}(x) = 2.$$

$$\Rightarrow df(X_t) = d(X_t^2) = 2X_t dX_t + \underbrace{dX_t dX_t}_{= \sigma^2 dt}$$

$$\begin{aligned} \Rightarrow df(X_t) &= d(X_t^2) = 2X_t(\mu dt + \sigma dW_t) + \sigma^2 dt \\ &= (2X_t\mu + \sigma^2)dt + 2X_t\sigma dW_t \end{aligned}$$

2. $f(x) = e^x$

$$f_t(x) = 0, f_x(x) = e^x, f_{xx}(x) = e^x$$

$$\begin{aligned} (*) \Rightarrow df(X_t) &= d(e^{X_t}) = e^{X_t} dX_t + \frac{1}{2}e^{X_t} \underbrace{dX_t dX_t}_{= \sigma^2 dt} \\ &= e^{X_t}(\mu dt + \sigma dW_t) + \frac{1}{2}e^{X_t}\sigma^2 dt \\ &= e^{X_t}(\mu + \frac{1}{2}\sigma^2)dt + e^{X_t}\sigma dW_t \end{aligned}$$

3. $f(x) = \sin(x)$

$$f_t(x) = 0, f_x(x) = \cos(x), f_{xx}(x) = -\sin(x)$$

$$\begin{aligned} (*) \Rightarrow df(X_t) &= d(\sin(X_t)) = \cos(X_t) dX_t - \frac{1}{2}\sin(X_t) \underbrace{dX_t dX_t}_{= \sigma^2 dt} \\ &= \cos(X_t)(\mu dt + \sigma dW_t) - \frac{1}{2}\sin(X_t)\sigma^2 dt \\ &= (\cos(X_t)\mu - \frac{1}{2}\sin(X_t)\sigma^2)dt + \cos(X_t)\sigma dW_t \end{aligned}$$

Exercise 2:

Let $X_t = W_t$.

1. Apply Itô's formula to compute

$$d(e^{aW_t - \frac{1}{2}a^2t})$$

2. Show that the process $M_t = e^{aW_t - \frac{1}{2}a^2t}$ is a martingale.

1. $f(t, x) = e^{ax - \frac{1}{2}a^2t}$

$$f_t(t, x) = -\frac{1}{2}a^2 e^{ax - \frac{1}{2}a^2t} = -\frac{1}{2}a^2 f(t, x)$$

$$f_x(t, x) = a e^{ax - \frac{1}{2}a^2t} = a f(t, x)$$

$$f_{xx}(t, x) = a^2 e^{ax - \frac{1}{2}a^2t} = a^2 f(t, x)$$

$$(*) \Rightarrow df(t, X_t) = df(t, W_t) = -\frac{1}{2}a^2 f(t, W_t) dt$$

$$\parallel \\ dM_t$$

$$+ a f(t, W_t) dW_t$$

$$+ \frac{1}{2} a^2 f(t, W_t) \underbrace{dW_t dW_t}_{=dt}$$

$$= a f(t, W_t) dW_t = dt$$

$$= a e^{aW_t - \frac{1}{2}a^2t} dW_t$$

$$\Rightarrow dM_t = a M_t dW_t$$

2. The Itô-integral $I(t) = \int_0^t dM_u dW_u = \int_0^t a M_u du$ is a martingale:

For $0 \leq s \leq t$:

$$\mathbb{E}[I(t) - I(s) | \mathcal{F}_s] = 0.$$

$$\mathbb{E}\left[\int_s^t a M_u du | \mathcal{F}_s\right] = 0.$$

So for $dM_t = a M_t dW_t$

$$\Rightarrow M_t = M_0 + \int_0^t a M_u dW_u = \overbrace{(M_0 + \int_0^s a M_u dW_u)}^{M_s} + \int_s^t a M_u dW_u$$

$$\text{Then, } \mathbb{E}[M_t | \mathcal{F}_s] = M_s + \mathbb{E}\left[\int_s^t a M_u dW_u | \mathcal{F}_s\right]$$

$$= \mathbb{E}[I(t) - I(s) | \mathcal{F}_s] = 0$$

$$\Rightarrow \mathbb{E}[M_t | \mathcal{F}_s] = M_s$$

So, M_t is a martingale.

(Or, shorter):

$$\text{Since } dM_t = a M_t dW_t,$$

there is no dt -term $\Rightarrow$ zero drift

$\Rightarrow M_t$ martingale.

Exercise 3: Let $X_t = W_t$.

1. Use Itô's formula to compute $d(W_t^3)$.
2. Find a constant c such that the process $M_t = W_t^3 - ctW_t$ is a martingale.

1.

$$f(x) = x^3 \\ f_x(x) = 3x^2, \quad f_{xx}(x) = 6x$$

$$\begin{aligned} (*) \Rightarrow d(X_t^3) &= d(W_t^3) = 3W_t^2 dW_t + \frac{1}{2} 6W_t \underbrace{dW_t dW_t}_{dt} \\ &= 3W_t^2 dW_t + 3W_t dt \end{aligned}$$

2. $dM_t = d(W_t^3) - c d(tW_t)$

$$d(tW_t) = ? \quad \begin{aligned} g(t, x) &= tx \\ g_t(t, x) &= x \\ g_x(t, x) &= t \\ g_{xx}(t, x) &= 0 \end{aligned}$$

$$(*) \Rightarrow d(tW_t) = W_t dt + t dW_t$$

$$\Rightarrow dM_t = (3W_t^2 - ct) dW_t + (3W_t - cW_t) dt$$

For a martingale, the drift must be zero

$$\Rightarrow 3W_t - cW_t = 0 \Leftrightarrow c = 3$$

Exercise 4: Let X_t satisfy

$$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t.$$

1. Apply Itô's formula to $Y_t = e^{\kappa t} X_t$.
2. Deduce an explicit representation for X_t .

1.

$$\begin{aligned} f(t, x) &= e^{\kappa t} x \\ f_t(t, x) &= \kappa e^{\kappa t} x \\ f_x(t, x) &= e^{\kappa t} \\ f_{xx}(t, x) &= 0 \end{aligned}$$

$$\begin{aligned} (*) \Rightarrow dY_t &= \kappa e^{\kappa t} X_t dt + e^{\kappa t} dX_t \\ &= \cancel{\kappa e^{\kappa t} X_t dt} + e^{\kappa t} \cancel{\kappa(\theta - X_t)dt} + e^{\kappa t} \sigma dW_t \\ &= e^{\kappa t} \kappa \theta dt + e^{\kappa t} \sigma dW_t \end{aligned}$$

2.

$$\Rightarrow Y_t = \underbrace{Y_0}_{=X_0} + \kappa \theta \int_0^t e^{\kappa s} ds + \sigma \int_0^t e^{\kappa s} dW_s$$

$$\int_0^t e^{\kappa s} ds = \frac{e^{\kappa t} - 1}{\kappa}$$

$$\Rightarrow Y_t = X_0 + \theta(e^{\kappa t} - 1) + \sigma \int_0^t e^{\kappa s} dW_s$$

$$X_t = e^{-\kappa t} Y_t \Rightarrow X_t = X_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t}) + \sigma e^{-\kappa t} \int_0^t e^{\kappa s} dW_s$$

Exercise 5

Let $u(t, x) = t^2 \cos x$.
Apply Itô's formula to compute $du(t, W_t)$.

$$u_t(t, x) = 2t \cos x$$

$$u_x(t, x) = -t^2 \sin x$$

$$u_{xx}(t, x) = -t^2 \cos x$$

$$dW_t dW_t = \overbrace{dt}^1$$

$$\begin{aligned} (*) \Rightarrow du(t, W_t) &= 2t \cos(W_t) dt - t^2 \sin(W_t) dW_t - \frac{1}{2} t^2 \cos(W_t) \overbrace{dt}^1 \\ &= (2t - \frac{1}{2} t^2) \cos(W_t) dt - t^2 \sin(W_t) dW_t. \end{aligned}$$

Exercise 6

Find a constant a such that
 $M_t = e^{at} \cos(W_t)$ is a martingale.

$$f(t, x) = e^{at} \cos x$$

$$f_t(t, x) = a e^{at} \cos x$$

$$f_x(t, x) = -e^{at} \sin x$$

$$f_{xx}(t, x) = -e^{at} \cos x$$

$$dW_t dW_t = \overbrace{dt}^1$$

$$\begin{aligned} (*) \Rightarrow dM_t &= a e^{at} \cos(W_t) dt - e^{at} \sin(W_t) dW_t - \frac{1}{2} e^{at} \cos(W_t) \overbrace{dt}^1 \\ &= (a - \frac{1}{2}) e^{at} \cos(W_t) dt - e^{at} \sin(W_t) dW_t \end{aligned}$$

If the drift is zero, then it is a martingale.

$$\Rightarrow (a - \frac{1}{2}) e^{at} \cos(W_t) = 0 \Rightarrow a = \frac{1}{2}$$